

Concepts: from instances to meaning



16-721: Learning-Based Methods in Vision
A. Efros, CMU, Spring 2009

Understanding an Image: what do we mean?



Recognizing Exact Instances?



A Beijing City Transit Bus #17, serial number 43253?

"It irritated him that the "dog" of 3:14 in the afternoon, seen in profile, should be indicated by the same noun as the dog of 3:15, seen frontally."

"My memory, sir, is like a garbage heap"

Jorge Luis Borges

Fumes the Memorious

Need more general (useful) information



What can we say the very first time we see this thing?

Functional:

- A large vehicle that may be moving fast, probably to the right, and will kill you if you stand in its way.
- However, at specified places, it will allow you to enter it and transport you quickly over large distances.

Communicational:

- bus, autobus, λεωφορείο, ônibus, автобус, 公共汽车, etc.

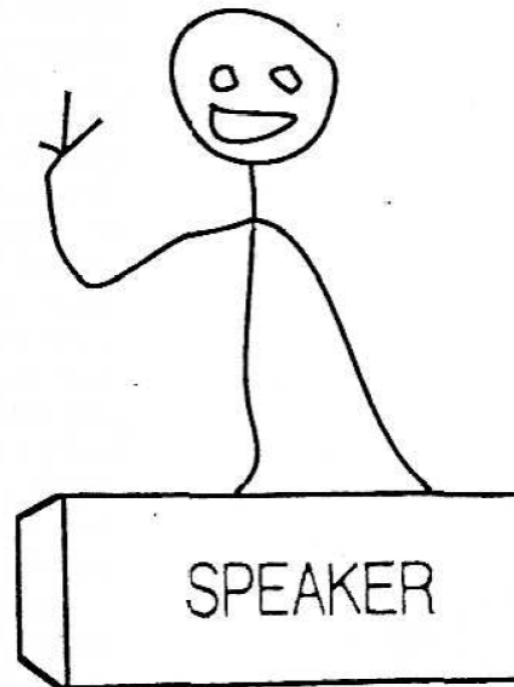
Concepts try to reduce complexity

Functional:

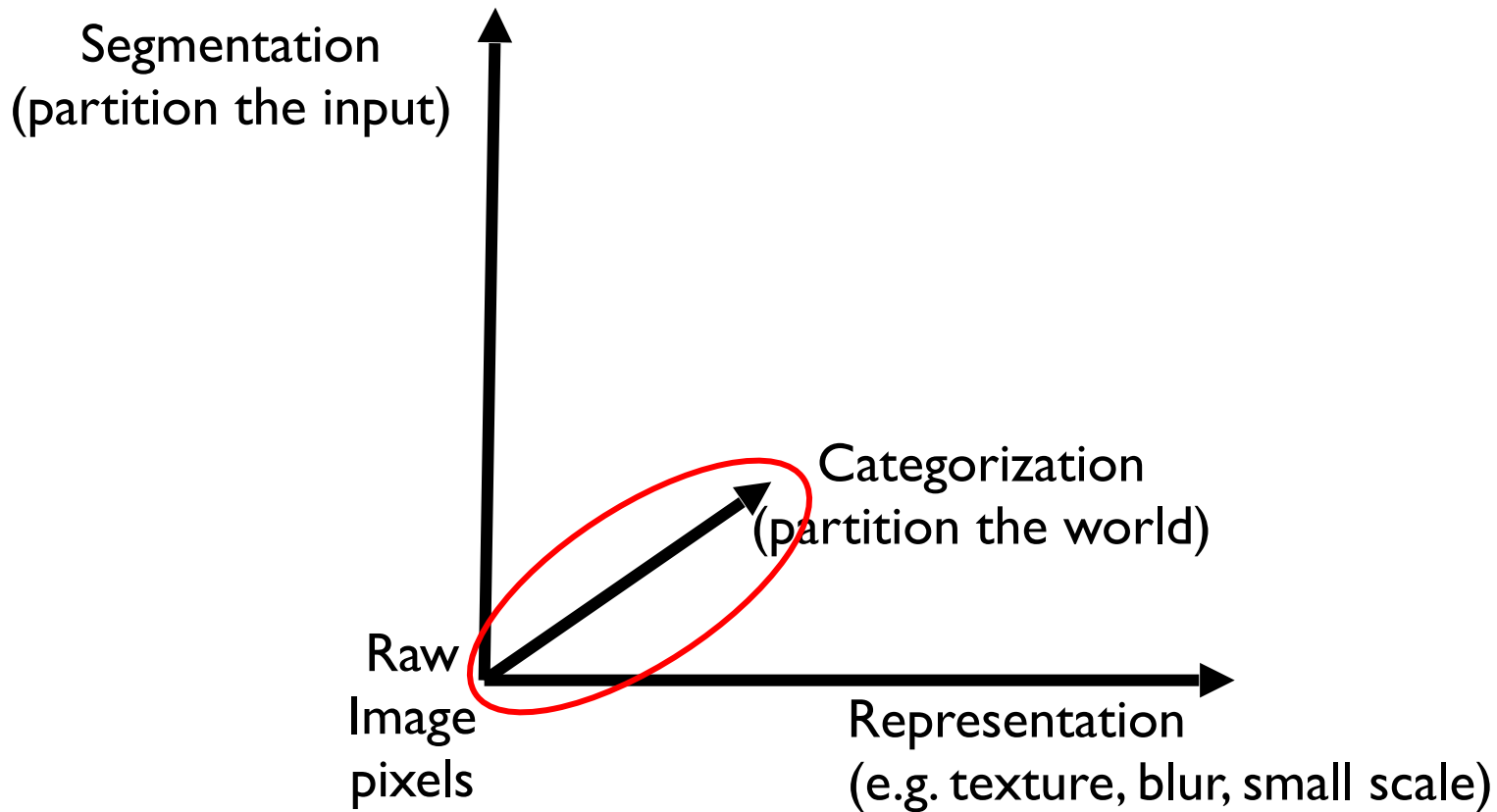
- Many instances act/behave in similar ways. If one tiger ate your cousin, then another tiger might very well eat you.

Communicational:

- There are way more object instances in the world than we have names for.



Ways of Reducing Complexity



Most of computer vision right now focuses on
“communicational” categorization...

Object naming -> Object categorization



Object categorization

*A picture is worth a 1000 words...
Or just 10?*

sky

building

flag

face

banner

wall

street lamp

bus

bus

cars

But it's all about function!

Let's downplay “Communicational” reasons.

They don't have strong connections to vision and might confuse our discussion, e.g.

- “Women, Fire, and Dangerous Things” is a category in Australian aboriginal language (Lakoff 1987)

Two approaches

Affordances

Perception of
Physical Structure

Flat surface
Horizontal
Knee-high
(etc.)

Perception of
Affordances

Sittable upon

Categorization

Perception of
Physical Structure

Flat surface
Horizontal
Knee-high
(etc.)

Categorization

Chair

Retrieval of
Function

Sittable upon

Affordances

Functions of an object that an observer can perceive directly from its visible structure.



throwable



sittable-upon



drinkable-from

Gestaltists again...

To primitive man each thing says what it is and what he ought to do with it: a fruit says, "Eat me"; water says, "Drink me"; thunder says, "Fear me," and woman says, "Love me."

-- Kurt Koffka

Comments on Affordances:

Interesting ideas:

Function follows form

Observer relativity

Similar to Gestalt idea of “physiognomic character”

Problems



Won't work for everything



Functional fixedness



Exaggerated claims

Categorization:

The process of perceiving objects as members of known types to allow observers to respond appropriately via past experiences stored in memory.

Four components of categorization:

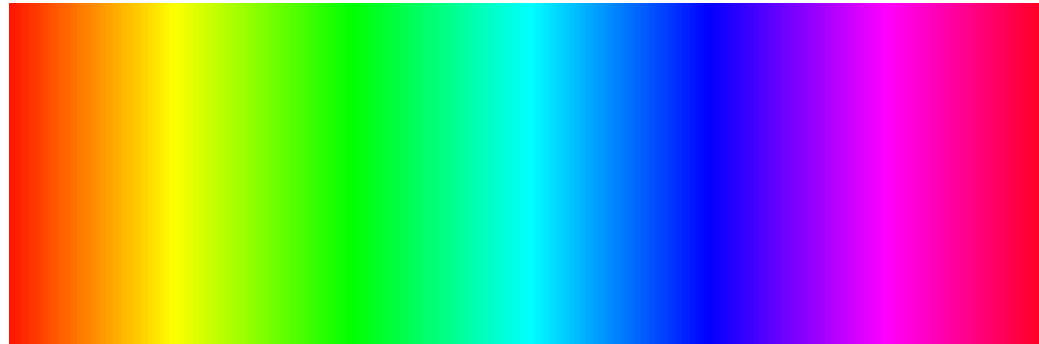
1. Representation of object (from the visual system)
2. Representation of categories (from memory)
3. Comparison process between 1 and 2
4. Decision process

An example of categorical perception

- Continuous perception: graded response



- Categorical perception: “sharp” boundaries

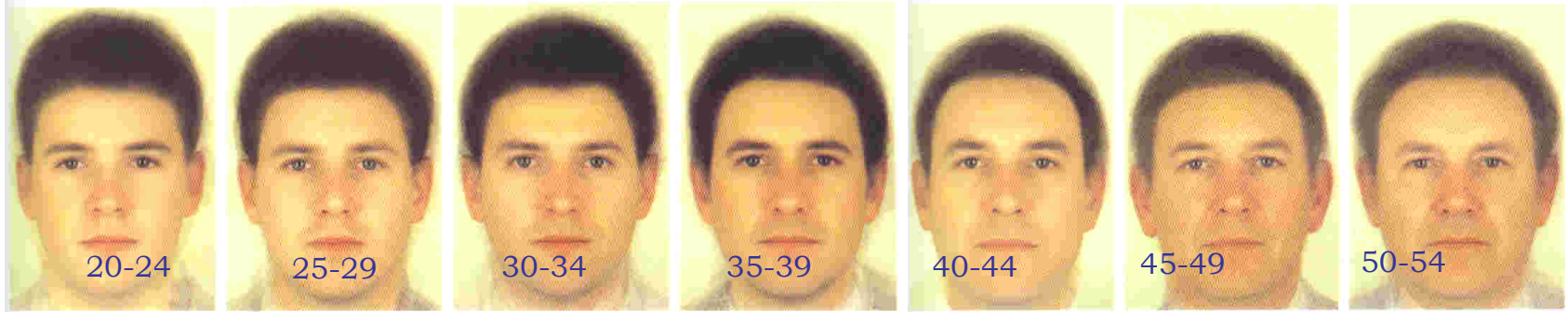


Many perceptual phenomena are a mixture of the two: categorical at an everyday level of magnification, but continuous at a more microscopic level. It can also depend on cultural aspects, expertise, task, attention, ...

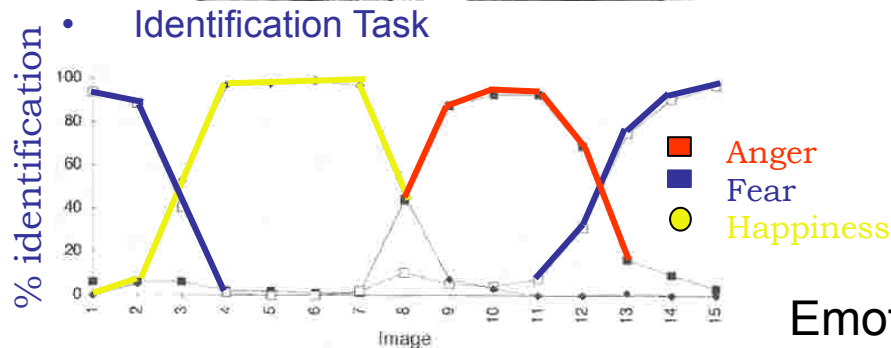
Slide by Torralba

Another example

- Continuous perception: graded response



- Categorical perception: “sharp” boundaries



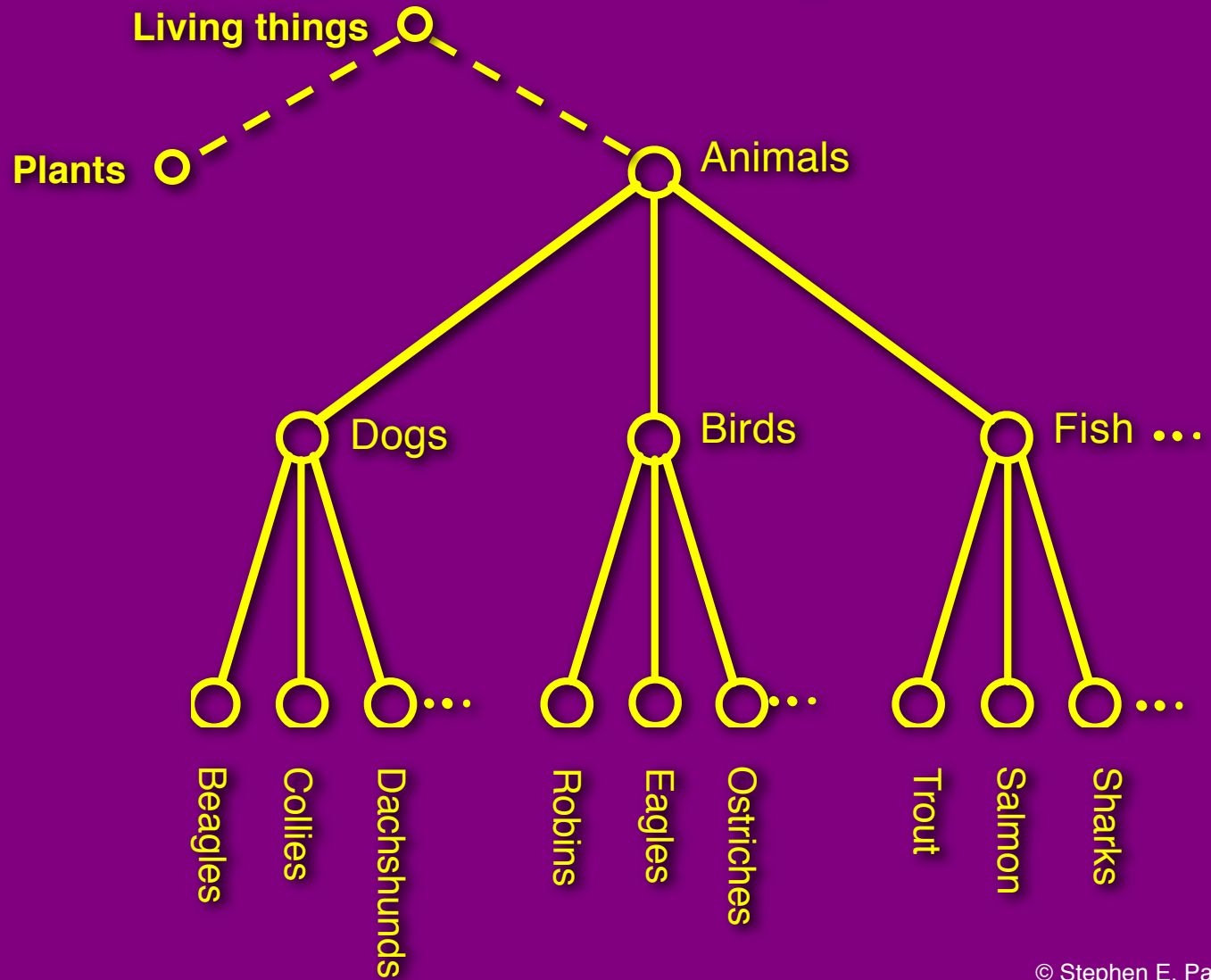
Slide by Torralba
Emotions have categorical boundaries

Classical View of Categories

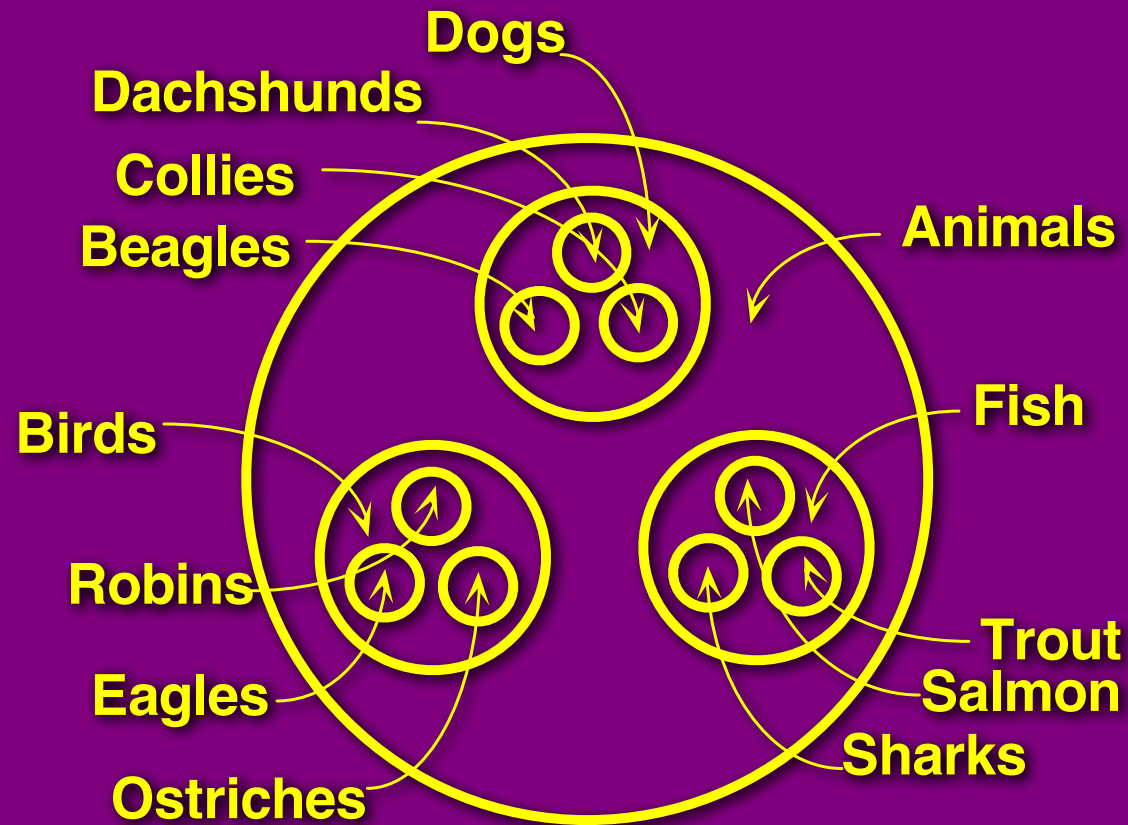
- Dates back to Plato & Aristotle
 1. Categories are defined by a list of properties shared by all elements in a category
 2. Category membership is binary
 3. Every member in the category is equal



Multiple Levels of Categories



Venn diagrams of categorical hierarchies



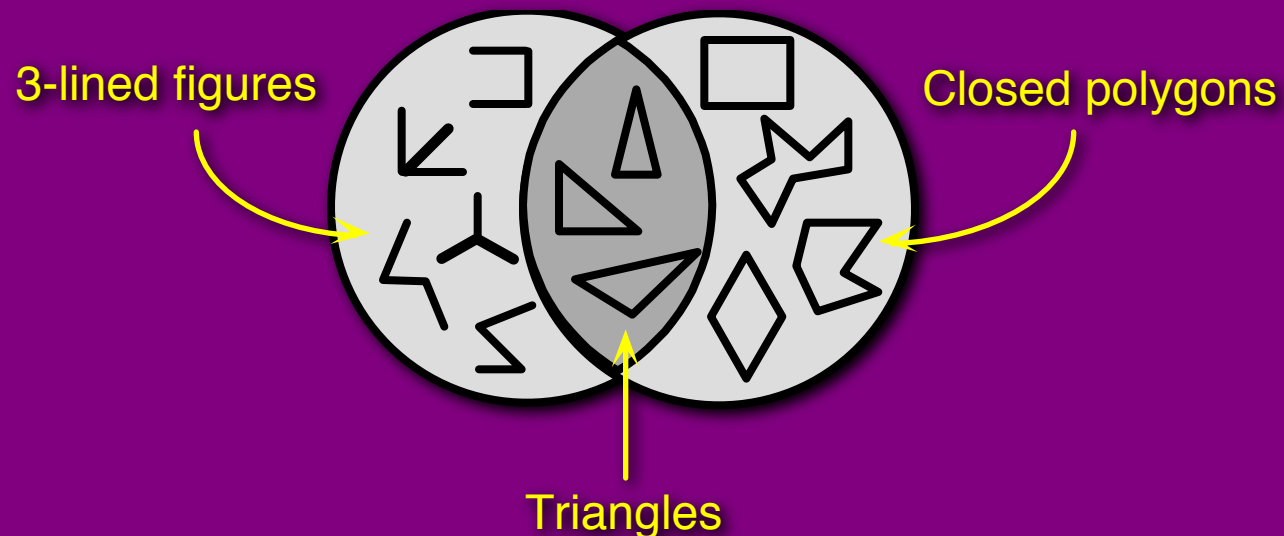
Aristotelian categories

Defined by necessary and sufficient conditions

Crisp boundary conditions

All members are equal

Example: Triangles are three-sided closed polygons



Problems with Classical View

- Humans don't do this!
 - People don't rely on abstract definitions / lists of shared properties (Rosch 1973)
 - e.g. Are curtains furniture?
 - Typicality
 - e.g. Chicken -> bird, but bird -> eagle, pigeon, etc.
 - Intransitivity
 - e.g. car seat is chair, chair is furniture, but ...

It gets worse!

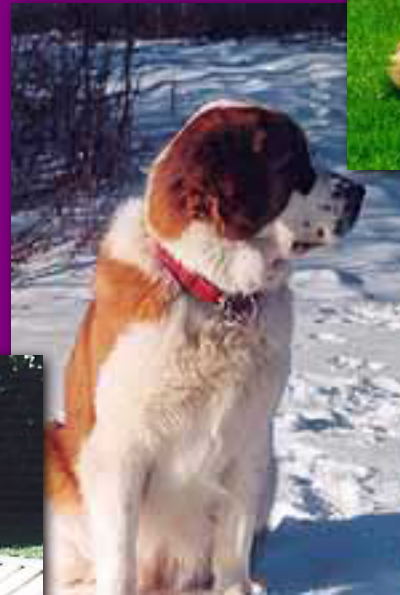
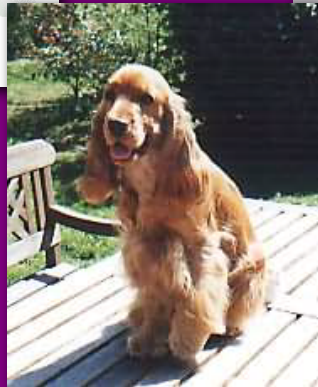
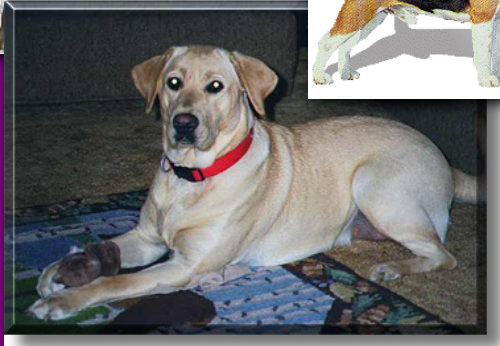
- Multiple category membership (it's not a tree, it's a forest!)
 - e.g. Tolstoy's "War and Peace" belongs to:
 - love story
 - Napoleonic wars
 - long Russian novels with lots of French dialog
- Doesn't work even in human-defined domains
 - e.g. Is Pluto a planet?

Natural categories (according to Rosch)

Defined by best examples (prototypes)

Graded membership function

Fuzzy boundary conditions



Evidence for prototypes

Typicality ratings

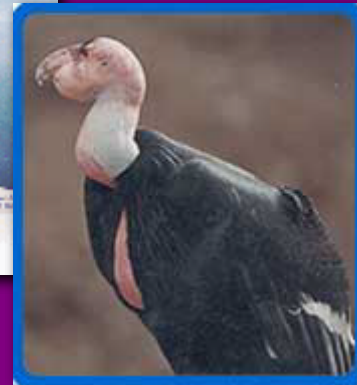
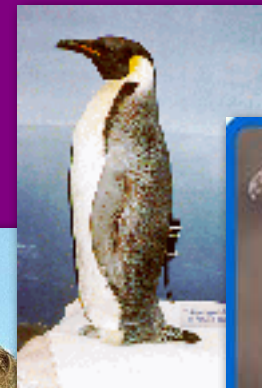
(How good are robins as an example of birds)

Production order of exemplars

(Name all the kinds of birds you can think of)

Time to verify categorical statements

(True or false: A robin is a bird)



Basic Level (Rosch)

A privileged intermediate level of the categorical hierarchy as defined by three operational criteria:

Shape Similarity: highest level at which members have similar shapes (e.g., dogs, not animals).

Similar motor interactions: highest level at which we interact with exemplars in the same way (e.g., pianos, not musical instruments).

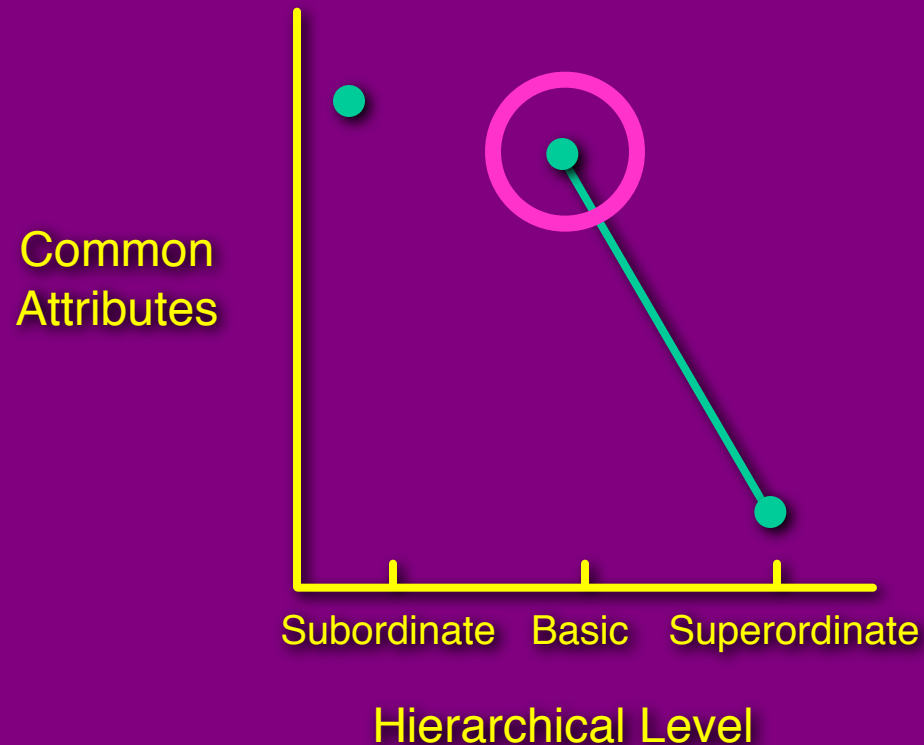
Common attributes: highest level at which members have the same features (e.g., chairs, not furniture).

Criteria for Basic Level Categories

Shape Similarity

Similar motor interactions

Common attributes



Are objects initially categorized at the basic level?

Jolicoeur, Gluck & Kosslyn (1984) asked subjects to name objects with the first label that came to mind.

Typical Exemplars



Bird, not Robin



Bird, not Sparrow



Bird, not Bluejay

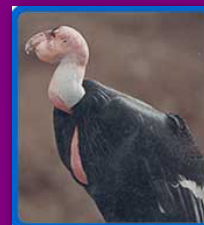
Atypical Exemplars



Ostrich, not Bird



Penguin, not Bird



Vulture, not Bird

Entry level categories

The level at which objects are first categorized perceptually.
Higher level categorization is conceptual.
Lower level categorization requires further perception.

Basic Level

“Bird” is the basic level category for every bird

Entry Level

“Bird” is the entry level category for typical birds,
but subordinate categories are the entry level for
atypical birds.

Canonical Perspective

The “best,” most easily identified view of an object.
(Palmer, Rosch & Chase, 1981)



All views of the horse



BEST (1.60)



SIDE (1.84)



FRONT-SIDE (2.12)



FRONT-SIDE-TOP (2.80)



SIDE-TOP (3.48)



FRONT (3.72)



BACK-SIDE (4.12)



BACK-SIDE-TOP (4.29)



FRONT-TOP (4.80)



BACK-TOP (5.56)

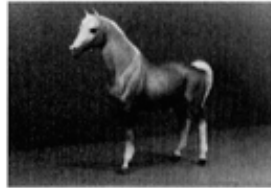


BACK (5.68)



TOP (6.36)

Canonical perspectives of all objects



HORSE



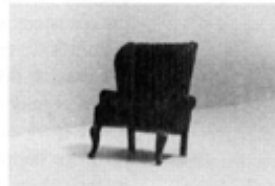
PIANO



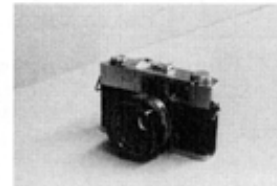
TEAPOT



CAR



CHAIR



CAMERA



CLOCK



TELEPHONE



HOUSE



PENCIL SHARPENER



SHOE



IRON

why?

- Frequency hypothesis
- Maximum Information hypothesis

We do not need to recognize the exact category

- A new class can borrow information from similar categories



Prototype or Sum of Exemplars ?

- Prototype Model

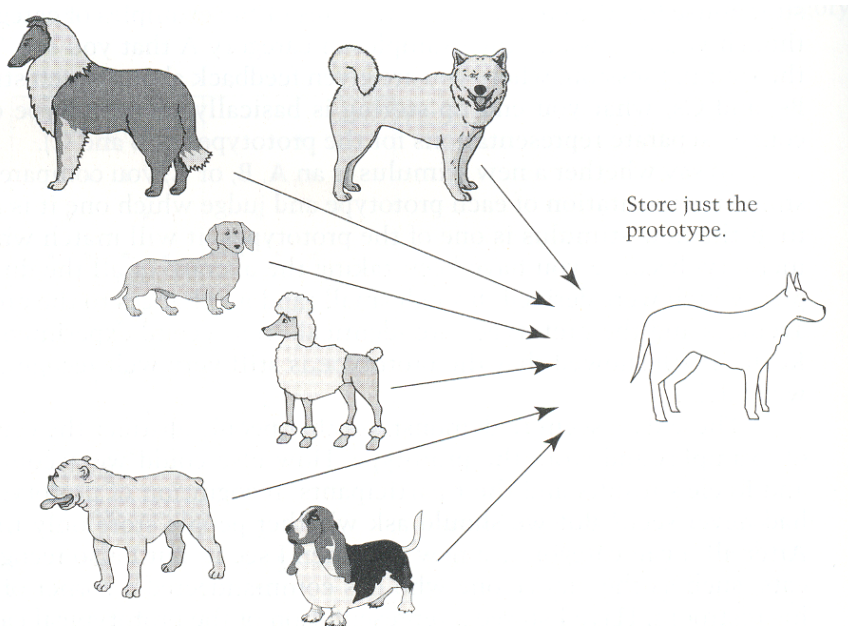


Figure 7.3. Schematic of the prototype model. Although many exemplars are seen, only the prototype is stored. The prototype is updated continually to incorporate more experience with new exemplars.

- Category judgments are made by comparing a new exemplar to the prototype.

- Exemplars Model

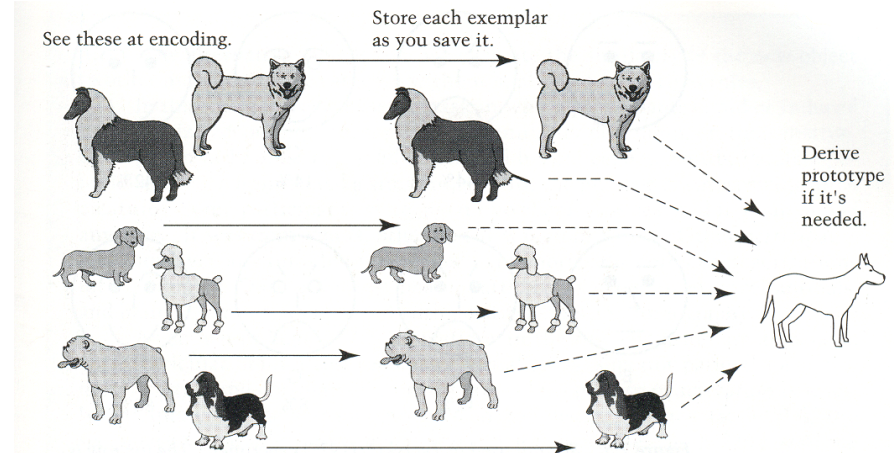


Figure 7.4. Schematic of the exemplar model. As each exemplar is seen, it is encoded into memory. A prototype is abstracted only when it is needed, for example, when a new exemplar must be categorized.

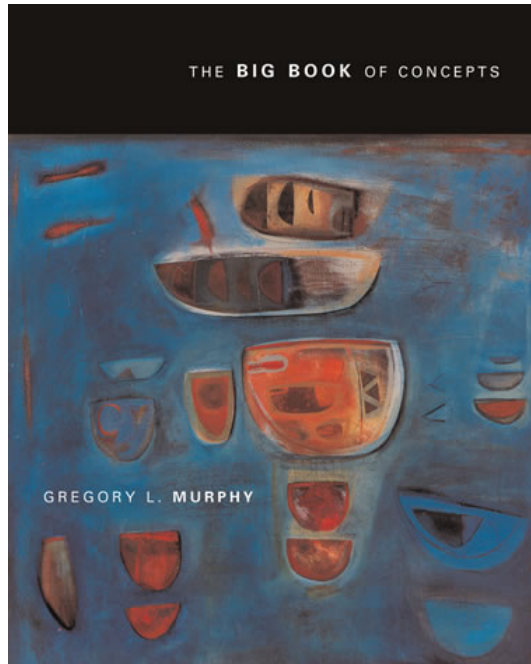
- Category judgments are made by comparing a new exemplar to all the old exemplars of a category or to the exemplar that is the most appropriate

Could be the same thing...

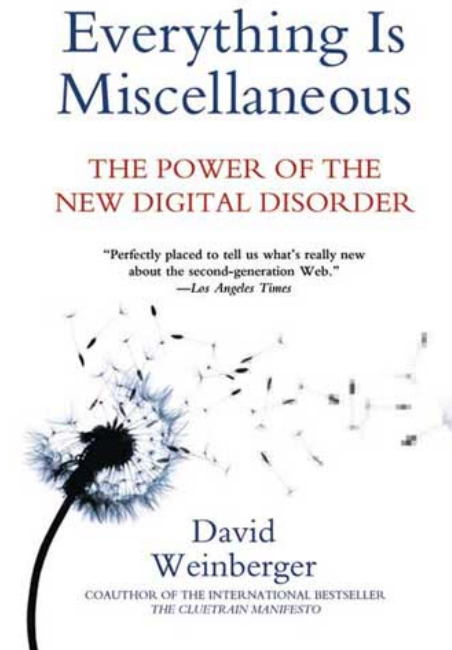


Think of
visual
“memex”

Further Reading



Murphy
Big Book of Concepts



Weinberger
Everything is Miscellaneous